MISH 21

CRANE SUSPENDED WORK PLATFORMS

AGC CSEA Alignment

Task Design - Engineering Controls

OSHA Core Element

Hazard Prevention and Control

Section Summary

Establishes requirements for crane-suspended work platforms including engineering design, load testing, and operational procedures.

Core Values: Integrity | Honesty | Professionalism | Thoroughness

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CRANE SUSPENDED WORK PLATFORMS

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AGC CSEA Alignment	OSHA Core Element
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1.0 PURPOSE

To define the responsibilities and requirements for protecting personnel engaged in the use of crane-suspended personnel platforms, and to ensure compliance with Occupational Safety and Health Administration/Washington Industrial Safety and Health Act (OSHA/WISHA) requirements.

2.0 RESPONSIBILITY

2.1 MH Construction Industrial Safety and health (IS&H) shall review and approve all uses of crane suspended personnel platforms.

2.2 The Key Supervisor shall be responsible for the following:

2.2.1 Ensuring employee compliance with this procedure.

2.4 Employees shall be responsible for complying with this procedure.

3.0 DEFINITIONS

Qualified Person: One who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training, and experience, has successfully demonstrated his/her ability to solve or resolve problems relating to the subject matter, the work, or the project.

4.0 GENERAL REQUIREMENTS

4.1 Crane-suspended work platforms shall comply with the following statutory requirements:

- 29 Code of Federal Regulations (CFR) 1926, Subpart N; Cranes, Derricks, Hoists, Elevators and Conveyors
- Washington Administrative Code (WAC) 296-155, Part L; Cranes, Derricks, Hoists, Elevators and Conveyors
- Hanford Hoisting and Rigging Manual

4.2 PRE-JOB REQUIREMENTS

4.2.1 The use of a crane to hoist employees on a personnel platform shall be prohibited, except when conventional means of reaching the work location present a greater hazard or are not possible because of structural design or worksite conditions. The Key Supervisor and MH Construction IS&H shall make a case-by-case evaluation to determine if an alternate method can be used. A justification for the use of crane-suspended personnel platforms shall be written into the Job Hazard Analysis (JHA) and approved by the MH Construction IS&H.

4.2.2 The crane to be used shall have been inspected within the past month. If the crane was out of service, it shall receive a complete annual inspection prior to placement in service.

4.2.3 The crane operator shall conduct a daily documented inspection prior to the start of work each shift.

4.2.4 Cranes with telescoping booms shall be equipped with a device to indicate clearly to the operator at all times the boom-extended length, or an accurate determination of the load radius to be used during the lift shall be made prior to hoisting personnel. A positive acting device shall be used which prevents contact between the load block or overall ball and the boom tip (anti-two-blocking device). This device shall deactivate the hoisting action before damage occurs. Friction type cranes shall be equipped with a two-block warning device.

4.2.5 A meeting attended by the crane operator, signal person, person(s) to be lifted, and the person responsible for the task to be performed shall be held to review work procedures, including the JHA. This meeting shall be held prior to the "trial lift" at each new work location, and shall be repeated for any employees newly assigned to the operation.

4.2.6 Prior to lifting employees, the operator and signal person shall conduct a full-cycle "trial lift" with an unoccupied personnel platform loaded to the anticipated lift weight. The operator shall determine that all systems, controls, and safety devices are functioning properly. The trial lift shall be repeated prior to hoisting employees whenever the crane is moved and set up in a new location, or returned to a previously used location. A visual inspection of the crane, rigging, and the personnel platform shall be conducted by the designated qualified person (and others at his/her direction) immediately after the trial lift to determine whether the trial lift has exposed any defective component or structure. Any defects found shall be corrected before hoisting personnel.

4.2.7 The "trial lift" process shall be documented on the Pre-lift Checklist (attachment MISH 21-A).

4.2.8 All newly constructed, modified, or repaired personnel platforms shall be load tested to 125 percent of rated capacity. Modified or repaired platforms shall be re-tested prior to use.

4.2.9 Tag lines shall be used unless their use creates an unsafe condition.

4.2.10 All lifts shall be made in accordance with the manufacturer's lifting recommendations.

4.2.11 The number of employees to be hoisted shall be kept to a minimum, and in no case shall it exceed four persons.

4.2.12 Employees working on the platform shall be considered as weighing at least 250 pounds each.

4.2.13 The platform shall not be used during high winds, electrical storms, snow, or other adverse weather conditions, which could endanger employees on the platform.

4.3 WORK PLATFORM REUIREMENTS

4.3.1 The personnel platform and suspension system shall be designed by a qualified engineer. The suspension system shall be designed to minimize tipping of the platform due to the movement of employees occupying the platform. The platform shall be capable of supporting its own weight and five times the maximum intended load without failure.

4.3.2 The personnel platform shall be conspicuously posted with the following information:

- Weight of the empty platform
- Maximum number of personnel
- Rated load capacity

4.3.3 The platform shall have standard railings as specified by OSHA/WISHA. The sides of the platform shall be enclosed from floor to mid-rail with either solid construction or expanded metal having openings no greater than ½ inch.

4.3.4 If access gates are installed, they shall swing inward and shall be equipped with a restraining device to prevent accidental opening.

4.3.5 The platform shall have overhead protection when employees are potentially exposed to falling objects.

4.3.6 The platform shall be easily identifiable by high visibility color

4.3.7 All welding of the personnel platform and its components shall be performed by a certified welder familiar with the weld grades, type, and material specified in the platform design.

4.4 BOX-TYPE PLATFORMS: In addition to the requirements of section 4.2, the following applies to box-type platforms:

4.4.1 Lifting bridles on box-type work platforms shall consist of four legs of equal length. One end shall be securely attached to each corner of the platform and the other end securely attached to a common ring, shackle, or other equivalent device to accommodate the crane hook or a strap to the crane hook.

4.4.2 The work platform shall be provided with an inside grabrail.

4.4.3 Each occupant shall wear a body harness in accordance with WISHA, and the lanyard shall be secured to a structural member capable of supporting a fall impact for employees using the anchorage point.

4.4.4 A substantial safety line shall pass through the eye of each leg of the bridle and shall be securely fastened with a minimum amount of slack to the lift line above the headache ball, or to the crane hook itself if a load block is in use (see attachments MISH 21-B and MISH 21-C).

4.4.5 The platform shall be constructed with a minimum five-to-one safety factor. Five hundred pounds shall be allowed for the first person with light tools, with at least an additional 250 pounds for each person added.

4.5 OPERATING REQUIREMENTS

4.5.1 The lift shall be performed under controlled conditions, and under the direction of an appointed signal person.

4.5.2 Communication between crane operator, signal person, and person(s) being lifted shall be maintained through standard hand signals or voice communication equipment. If radio equipment is used, it shall be equipped with a discrete channel.

4.5.3 The crane operator shall remain at the controls at all times when personnel are on the platform.

4.5.4 Lifting or lowering shall not exceed 100 feet (30.48 m) per minute. Downward motion shall always be with power or controlled lowering (no free fall).

4.5.5 Cranes shall not travel with personnel on the platform

4.5.6 The employee(s) being lifted, moved, or positioned shall remain in continuous sight or in contact with the operator or signal person.

4.5.7 Crane outriggers shall be fully extended and locked. A firm footing uniformly level within one percent shall be provided.

4.5.8 The total weight of the lifted load (including personnel) shall not exceed 50 percent of the rated capacity for the radius and configuration of the crane.

4.5.9 The crane hook shall be provided with a safety latch or other self-closing device.

5.0 PROCEDURE

Not Applicable.

6.0 RECORDS

Document

Record Submittal Responsibility Record Retention Responsibility

All Documents (generated by this procedure, e.g., design, welding, maintenance) Contractor MH Construction Pre-lift Checklist

Contractor MH Construction

7.0 REFERENCES

ANSI B30.2 Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings-Overhead and Gantry Cranes

8.0 ATTACHMENTS

MISH 21-A Pre-lift Checklist MISH 21-B Details of Safety Line Attachment MISH 21-C Multiple-Part Line block Suspension

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